Physical Chemistry Cume

Jan 27th 2012

Constants and conversion factors:

Useful integral:

$$\int_0^\infty x^n e^{-ax} dx = \frac{n!}{a^{(n+1)}}$$

1) The 1s atomic orbital (hydrogenic) wavefunction is given by the product of the spherical harmonic (angular) wavefunction and the radial wavefunction:

$$Y_{0,0} = \left(\frac{1}{4\pi}\right)^{1/2}$$

$$R_{1,0}(r) = 2\left(\frac{Z}{a_o}\right)^{3/2} e^{-\frac{Z \cdot r}{a_0}}$$

$$\psi_{1,0,0} = Y_{0,0} R_{1,0}(r) = \left(\frac{1}{4\pi}\right)^{1/2} \cdot 2\left(\frac{Z}{a_o}\right)^{3/2} e^{-\frac{Z \cdot r}{a_0}}$$

Where $\psi_{n,l,ml}$ is the total wavefunction, $Y_{l,ml}$ is the spherical harmonic (angular) wavefunction and $R_{n,l}(r)$ is is the radial wavefunction. *Z* is the nuclear charge and a_0 is the Bohr radius (52.9 pm = 5.29×10^{-11} m).

(a) What is the average distance for the 1s electron in Be^{3+}?

(40 points)

(b) If the 1s electron in Be^{3+} is excited to a state described by the wavefunction:

$$\psi_{4,1,1} = Y_{1,1} R_{4,1}(r)$$

Describe the transition using term symbols **(20 points)**

HINT: You do not need to know the actual wavefunction of $\psi_{4,1,1}$ to answer this

2) Which of the following electronic transitions are allowed? **(20 points)**

(i) $^3D_2 \rightarrow {}^3F_1$

(ii) $^3P_0 \rightarrow {}^3S_1$

(iii) $^3D_3 \rightarrow {}^1P_1$

(iv) $^3D_3 \rightarrow {}^1S_0$

(v) $^2P_{3/2} \rightarrow {}^2S_{1/2}$

3) How many total degenerate states does a ^3F term have? **(20 points)**